

SELECTED PROJECT 01

LLM Production Incident Assistant

From an alert to an
evidence-backed investigation.

PYTHON / FASTAPI / REACT / POSTGRESQL

100

synthetic benchmark cases

3

prepared demo scenarios

v2.2.0

verified release

THE CHALLENGE

Make incident reasoning inspectable.

On-call engineers must connect alerts with runbooks, previous incidents, and telemetry. This project brings those sources into a single workspace and links preliminary hypotheses to versioned evidence, while the engineer retains control over tool execution.

PROJECT SCOPE

Full-stack application, retrieval pipeline, tool controls, automated evaluation, and cloud delivery.

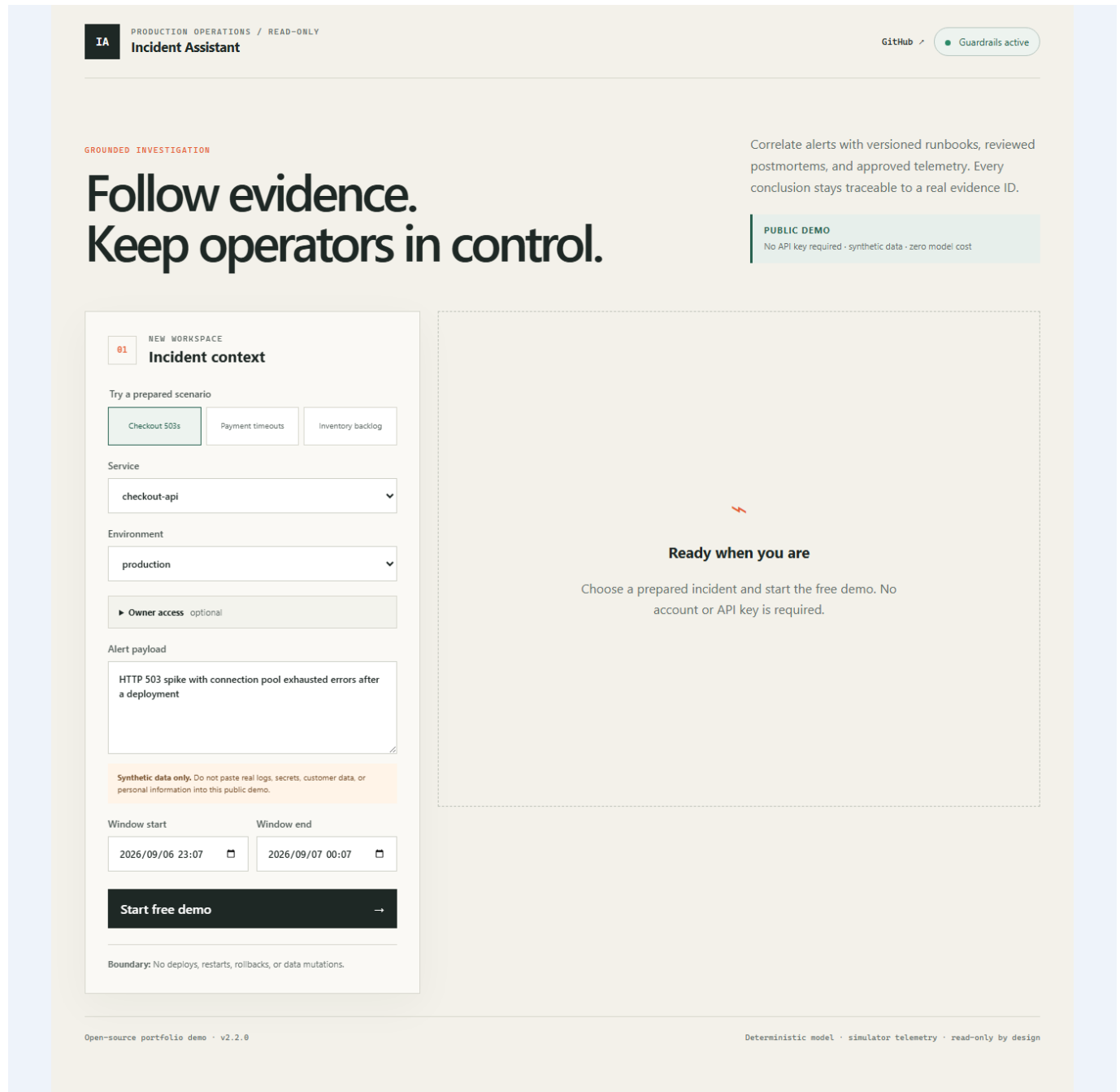
DELIVERY

Independent portfolio project.
Public source and a keyless demo.
Portfolio prepared September 11, 2026.

[Open live demo](#)[View source code](#)

A workflow an engineer can inspect.

Choose an incident. Follow the evidence. Review the proposed next queries.



Actual v2.2.0 entry workspace, captured during local browser verification. The public demo offers checkout, payments, and inventory scenarios.

7 evidence items

Versioned sources in the checkout smoke test

1 preliminary hypothesis

Assessment remains subject to operator review

3 pending queries

Deployment, log, and metric investigation

Grounding is a system concern.

The API connects retrieval, structured composition, and controlled tool execution.

01	Capture context Service, environment, alert, time window	React + TypeScript
02	Validate and route Schemas, API-key roles, public request limits	FastAPI + Pydantic
03	Retrieve and rank Full-text + hashed vectors; RRF, rerank, compress	PostgreSQL + pgvector
04	Compose the assessment Hypotheses and timeline reference returned evidence IDs	Structured model boundary
05	Review proposed tools Separate approval, scope checks, budgets, and audit	Read-only gateway

WHAT RUNS IN THE HOSTED DEMO

One Render web service; PostgreSQL/pgvector; deterministic generation; simulator telemetry; inline jobs. Visitors can investigate without a key. Operational routes retain API-key checks.

WHAT THE REPOSITORY ALSO SUPPORTS

A configurable Responses-compatible model with repair, fallback, cache, and cost tracking; fixed production telemetry adapters; Redis Queue workers; Docker Compose and Kubernetes deployment resources.

Design tradeoff: the public experience stays credential-free and reproducible. External models and real telemetry require a separately configured deployment and environment-specific validation.

Show the work. Bound the claims.

Recorded v2.2.0 verification, September 7, 2026. Source links are clickable.

CHECK	RECORDED RESULT
Local backend suite	40 passed; 1 Docker-dependent test skipped
Backend coverage	82.21% across measured packages
Synthetic evaluation	100 cases; strict and A/B gates passed
Desktop + mobile browser checks	4 Playwright checks passed
GitHub CI + release images	All four CI jobs; three image builds passed

HOW TO INTERPRET THE RESULTS

The benchmark uses an in-memory index and deterministic generation with 80 development and 20 held-out synthetic cases. Its proxy checks cover root-cause term matching, source hits, evidence-ID validity, and expected tools. They demonstrate regression behavior, not production accuracy or semantic proof.

LIMITATIONS THAT SHAPE THE NEXT ITERATION

Feature-hashed vectors are not trained semantic embeddings. Hypothesis confidence is heuristic. The shared demo corpus must stay synthetic; the application is not a tenant-isolation system. Free hosting can sleep, and the database has an expiry. Real-team use needs historical-incident evaluation and deployment-specific data access, retention, and backup decisions.

ENGINEERING VALUE

A working full-stack project that connects retrieval, evidence provenance, human-controlled tools, runtime configuration, and automated delivery into one inspectable incident workflow.

Source

CI evidence

Release images